# Utsav Amrish Joshi

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## Summary


- Mechanical engineer with plant floor experience in time studies, bottleneck analysis, and equipment performance validation.
- Built capacity and ROI models that shaped automation and expansion decisions at a high-volume food processing plant.
- Diagnosed mechanical, thermal, and electrical failure modes on industrial equipment through staged load testing and inspection.
- Comfortable running the data collection myself, then turning it into a documented recommendation someone can act on.


## Skills

**Tools:** Vibration & Alignment Testing, FAT/PDI Protocols, Pump Sizing
**Production:** Time Study & Bottleneck Analysis, Throughput Modeling, Capacity Planning, ROI/Payback Analysis, Process Mapping
**Quality:** GD&T, SQF, FDA/USDA Compliance, Audit Readiness, Lean Six Sigma, Root Cause Analysis
**Software:** MS Office (Excel, Outlook, Word, PowerPoint, Teams), SolidWorks, AutoCAD, Siemens NX, ANSYS, COMSOL
**Certifications:** CAD Certification (SolidWorks, AutoCAD, Siemens NX); Lean Six Sigma Yellow Belt - ASU

## Work Experience

### Manufacturing Engineering Intern | Shamrock Foods | Phoenix, Arizona, USA *05/2026 - 07/2026*

- Time-studied **4** shipping workflows (**2,551** cases/day); found **30%** of warehouse labor hours were automatable.
- Netted out **25.4 hr/day** of manual work automation keeps, leaving **22.6 labor-hr/day** a conveyor could remove.
- Built ROI/NPV model for an est. **$3.0M** sorting system; negative **15-yr** NPV at **8%** backed a "not yet" call.
- Benchmarked **11 lines** and **30+** machines vs. rated capacity; traced the gap to changeovers, not machines.
- Drew a 5-phase AutoCAD floor rebuild plan that kept production, forklift, and maintenance flows open.
- Tested a Copilot Studio QA agent (team of 4) over **65 docs**, now used in audits; cut **~15 min** lookups to **~1 min**.

### Manufacturing Engineer | Gangadhar Industries | Vapi, Gujarat, India *07/2023 - 04/2024*

- Ran ISO 8528 stepped load tests (**25 - 100%** plus **1-hr 110%** overload) on **5 - 625 kVA** single-bearing gensets.
- Caught **14.2 mm/s** RMS vibration on a **500 kVA** genset; fixed **0.18 mm** soft foot with operator to reach **3.6 mm/s**.
- Ran sudden-load transient tests on **5-58.5 kVA** units myself; supported the plant manager on larger sets.
- Logged coolant, exhaust, and Pt100 RTD winding temps using a thermocouple probe, IR pyrometer, and data logger.
- Checked soft foot, adapter face alignment, and mount/frame bolt torque marks before each load test.
- Recorded test data in FAT checklists and ran pre-delivery inspections before customer dispatch.

### Mechanical Engineer Intern | Linde Engineering | Vadodara, Gujarat, India *02/2023 - 04/2023*

- Calculated TDH by hand from the plot plan, P&ID and Mechanical / process data sheets, using Darcy-Weisbach for friction loss.
- Plotted system curves against catalogue pump curves for **6 pumps** and compared the options.
- Calculated NPSH available and compared it with NPSH required to check suction conditions.
- Reviewed results with my supervisor and repeated the selection with a larger pump as an alternative

## Education

### Master of Science, Mechanical Engineering *08/2024 - 05/2026*

Arizona State University, *Tempe, Arizona*

### Bachelor of Engineering, Mechanical Engineering *08/2020 - 06/2023*

Gujarat Technological University, *Ahmedabad, India*

## Projects

### Design of Experiment | Factorial Study | ASU *01/2025 - 04/2025*

- Ran a $2^3$ factorial DOE (16 randomized runs) in JMP; ANOVA showed ice mass drove ~95% of cooling-rate variation.
- Fit regression models ($R^2$ 0.94–0.97), checked residuals, and turned results into operating recommendations.

### Coupled Electromagnetic-Thermal Modeling of Microwave Heating | COMSOL | ASU *01/2026 - 08/2026*

- Built a coupled Maxwell-heat-transfer model in COMSOL (**2.45 GHz cavity, 250 mL water load**) for microwave heating.
- Validated hand-calculated penetration depth & energy balance; modeled permittivity via a **Debye relaxation** model.

### Transient Thermal Simulation - Additive Manufacturing | ANSYS, COMSOL | ASU *01/2025 - 04/2025*

- Modeled FDM layer-by-layer heat buildup in ANSYS (element activation) comparing ABS and PA12 heat-affected zones.
- Modeled laser powder-bed heating in COMSOL (Ti-6Al-4V); circular scans peaked at **~2405 K** vs **~2264 K** linear.

### Graduate Teaching Assistant | Arizona State University | MEE 351, MAE 545 *08/2025 - 05/2026*

- Communicated technical concepts via in-class COMSOL, ANSYS, and MATLAB demos, grading **100+** students.
- Redesigned the quiz and grading format for consistent, equitable outcomes across **170** students.